PORTFOLIO STRESS TESTING & SCENARIO ANALYSIS PROJECT

1. Introduction

Financial institutions face significant exposure to market risk due to fluctuations in asset prices. While traditional risk measures such as Value at Risk (VaR) provide useful estimates under normal market conditions, they often underestimate losses during extreme events—highlighted by the 2008 Global Financial Crisis.

To address these limitations, stress testing and scenario analysis are widely used to evaluate portfolio resilience under adverse conditions.

This project develops a comprehensive stress testing framework for a multi-asset portfolio using historical market data. It integrates:

- **Historical risk analysis**
- **Scenario-based stress testing**
- **Monte Carlo simulation**

The portfolio consists of four assets representing different market characteristics:

- **Apple Inc. (AAPL) – large-cap technology**
- **JPMorgan Chase & Co. (JPM) – banking sector**
- **Tesla Inc. (TSLA) – high-growth, high-volatility**
- **Vanguard FTSE Emerging Markets ETF (VWO) – emerging markets exposure**

The objective is to evaluate portfolio risk under both observed and simulated extreme conditions, using a fully reproducible pipeline implemented in R and SQL.

2. Data and Methodology

2.1 Data Extraction

Historical daily price data from January 2020 onward was sourced from Yahoo Finance using R. Data was stored in a MySQL database (financedata_db) to ensure:

- **Structured storage**
- **Efficient querying**
- **Reproducibility**

R was used for analysis, while SQL served as the data storage layer.

2.2 Data Processing

The dataset was structured in long format and transformed into a wide format for portfolio analysis.

Daily log returns were computed:

$$r_t = ln \frac{P_t}{P_{t-1}}$$

This approach ensures:

- **time additivity**
- **statistical consistency**

Missing values from lag operations were removed to maintain data integrity.

2.3 Portfolio Construction

An equally weighted portfolio was constructed:

- **AAPL: 25%**
- **JPM: 25%**
- **TSLA: 25%**
- **VWO: 25%**

Portfolio returns were computed as:

$$R_p = \sum w_i R_i$$

This provides a simple yet effective framework for analyzing diversification effects.

3. Exploratory Data Analysis

3.1 Price Dynamics

Asset prices exhibited:

- **clear bull and bear cycles**
- **sharp drawdowns (market shocks)**
- **varying volatility across assets**

Tesla showed the highest volatility, while Apple exhibited relatively stable growth.

3.2 Return Distributions Key findings:

- **Mean returns ≈ 0**
- **Presence of fat tails**
- **Significant variation in dispersion**

This indicates a higher probability of extreme events than predicted by a normal distribution.

3.3 Correlation Structure

The correlation matrix showed:

- **moderate positive correlations**
- **no perfect co-movement**

This suggests diversification benefits under normal conditions, though these may weaken during stress.

4. Risk Measurement

4.1 Portfolio Behavior

The portfolio return series exhibited:

- **reduced volatility compared to individual assets**

- **persistence of market-driven fluctuations**
- **occasional extreme movements**

4.2 Stylized Facts

The analysis confirmed key financial characteristics:

- **Volatility clustering**
- **Fat tails**
- **Asymmetric risk behavior**

These violate standard model assumptions and justify advanced risk modeling.

4.3 Risk Metrics

The following were computed:

- **Mean return**
- **Standard deviation**
- **Value at Risk (VaR)**
- **Expected Shortfall (ES)**

Key insight:

Expected Shortfall consistently exceeds VaR, highlighting tail risk that VaR alone fails to capture.

4.4 Base Case Risk Interpretation (Normal Market Conditions)

The portfolio's baseline risk profile provides critical insight into its behavior under non-stressed conditions.

The average daily return of approximately 0.078% suggests moderate positive performance, equivalent to an annualized return near 20%. However, this return is accompanied by relatively high volatility, with a daily standard deviation of 1.82% and an annualized volatility of approximately 29%, indicating a high-risk portfolio relative to typical market benchmarks.

The distribution of returns exhibits negative skewness (-0.63) and excess kurtosis (8.58). This implies that:

- **losses are more severe than gains (downside asymmetry), and**
- **extreme events occur more frequently than predicted by a normal distribution (fat tails).**

These characteristics confirm that the portfolio is exposed to significant downside tail risk, making it vulnerable to sudden market shocks.

From a risk measurement perspective:

- **The 1-day VaR (95%) of -2.75% (~$27,548) represents the expected loss threshold under normal conditions.**
- **However, the Expected Shortfall (95%) of -4.26% (~$42,607) indicates that when losses exceed VaR, they are substantially larger.**

The gap between VaR and ES highlights a key limitation of VaR:

it captures the cutoff point of losses but fails to reflect their magnitude in extreme scenarios.

Overall, the base case analysis demonstrates that even under normal conditions, the portfolio exhibits:

- **elevated volatility**
- **asymmetric risk exposure**
- **significant tail dependence**

This justifies the need for stress testing and advanced risk modeling techniques.

5. Scenario Analysis

Three stress scenarios were applied:

5.1 Market Crash (-20%)

The market crash scenario produced the most severe deterioration in portfolio performance, with significantly amplified downside risk and a pronounced left-tail shift in the return distribution. This indicates that the portfolio is highly exposed to systemic equity risk, where diversification fails as correlations converge during periods of market stress.

5.2 Interest Rate Shock

The interest rate shock scenario revealed asymmetric sector sensitivity, with growth-oriented assets exhibiting greater volatility compared to financial institutions. This suggests that portfolio risk is not uniformly distributed across assets but is influenced by macroeconomic factors such as monetary policy.

5.3 Emerging Market Crisis

The emerging market crisis scenario demonstrated that localized shocks can propagate through the portfolio via spillover effects. While diversification partially mitigates losses, exposure to emerging markets introduces additional vulnerability to global financial instability.

5.4 Stress Comparison

- **Market crash → highest risk**
- **Rate shock → moderate**
- **EM crisis → contained**

Key insight:

Diversification weakens significantly during systemic shocks.

5.5 VaR and ES Under Stress

The consistent divergence between Value at Risk and Expected Shortfall highlights the limitations of VaR as a standalone risk measure. Expected Shortfall provides a more comprehensive view of extreme losses by capturing tail risk beyond the VaR threshold.

This confirms tail risk amplifies disproportionately during crises.

6. Monte Carlo Simulation

A Monte Carlo simulation (10,000 scenarios) was performed using a multivariate normal distribution based on historical mean and covariance.

While Monte Carlo simulation provides a forward-looking distribution of potential outcomes, its reliance on the normality assumption may lead to an underestimation of extreme losses. This reinforces the importance of complementing simulation with stress testing.

Key Findings:

- **Simulated distribution confirms tail risk**
- **Monte Carlo VaR aligns with historical estimates**
- **Differences highlight model assumptions**

Critical Insight:

The assumption of normality may underestimate extreme losses, reinforcing the importance of stress testing.

6.1 Model Comparison: Normal vs Student-t VaR

To address the limitations of the normality assumption in Monte Carlo simulation, a multivariate Student-t distribution was implemented to better capture heavy tails observed in empirical returns.

Results

Model	VaR 95%	VaR 99%
Normal Monte Carlo	-2.95%	-4.14%
Student-t Monte Carlo	-3.63%	-5.59%

Key Insights

1. Student-t Captures Higher Tail Risk

The Student-t model produces consistently more negative VaR estimates:

- **~23% larger losses at 95%**
- **~35% larger losses at 99%**

This reflects its ability to model fat tails, unlike the normal distribution.

2. Underestimation of Risk Under Normality

The normal Monte Carlo VaR:

- **aligns closely with historical VaR**
- **but fails to capture extreme downside risk**

In contrast, the Student-t model:

- **produces more conservative (realistic) estimates**
- **better reflects observed kurtosis (~8.6) in the data**

3. Tail Risk is Model-Dependent

This comparison demonstrates that:

risk estimates are highly sensitive to distributional assumptions

Two models using the same data can produce materially different risk measures.

This is a critical concept in risk management:

- **model risk is real risk**

4. Implications for Risk Management

- **Using normal-based VaR may lead to systematic undercapitalization**
- **Student-t VaR provides a more robust buffer against extreme losses**
- **Regulatory frameworks (e.g., Basel) increasingly favor Expected Shortfall + stress testing for this reason**

The significant divergence between normal and Student-t VaR estimates confirms that the assumption of normally distributed returns leads to a systematic underestimation of tail risk. Incorporating fat-tailed distributions provides a more realistic representation of extreme market behavior, particularly for portfolios exhibiting excess kurtosis.

7. Discussion

7.1 Portfolio Risk Profile

The portfolio demonstrates a high-risk, growth-oriented structure, driven by exposure to technology, banking, and emerging markets assets. While the equal weighting strategy provides diversification benefits under normal conditions, the overall volatility (~29% annualized) remains elevated.

Importantly, the presence of negative skewness and excess kurtosis indicates that risk is not symmetrically distributed. Instead, the portfolio is characterized by rare but severe losses, a defining feature of financial market behavior.

7.2 Diversification and Its Limits

The correlation structure suggests that diversification is effective during stable market periods. However, scenario analysis reveals that:

- **correlations tend to increase during stress**
- **diversification benefits diminish significantly**

This is particularly evident in the market crash scenario, where losses approximately double relative to the base case.

This behavior reflects a well-documented phenomenon in financial markets:

"Diversification works until it is needed most."

7.3 Tail Risk and Model Limitations

The consistent divergence between Value at Risk (VaR) and Expected Shortfall (ES) highlights the presence of substantial tail risk.

While VaR provides a useful benchmark, it systematically underestimates:

- **the severity of extreme losses**
- **the probability of tail events**

Expected Shortfall, by contrast, captures the average loss beyond the VaR threshold, making it a more reliable measure in stress environments.

This distinction becomes more pronounced under stressed scenarios, where losses escalate disproportionately.

The comparison between normal and Student-t Monte Carlo simulations further reinforces the presence of heavy tails in the return distribution. The Student-t model produces significantly larger VaR estimates, particularly at the 99% level, indicating that extreme losses are more likely than predicted under normality.

This highlights the importance of selecting appropriate distributional assumptions when modeling financial risk, as incorrect assumptions can lead to substantial underestimation of potential losses.

While the Student-t distribution improves tail modeling, extreme value theory (EVT) provides a theoretically grounded framework for modeling the tail directly.

7.4 Time Horizon Effects

The analysis of VaR across multiple horizons reveals that risk scales non-linearly over time. For example:

- **1-day VaR ≈ 2.75%**
- **21-day VaR ≈ 12.6%**

This demonstrates that:

- **prolonged exposure significantly increases potential losses**
- **compounding effects amplify downside risk**

Under stressed conditions, this effect becomes even more pronounced, with potential drawdowns exceeding 25% of portfolio value over a one-month horizon.

7.5 Scenario Sensitivity and Risk Drivers

The stress scenarios highlight distinct sources of portfolio vulnerability:

- **Market Crash:**
  The most severe losses, driven by systemic equity risk and correlation breakdown.
- **Interest Rate Shock:**
  Uneven impact across sectors, reflecting sensitivity of growth assets to monetary tightening.
- **Emerging Market Crisis:**
  Evidence of spillover effects, where localized shocks propagate through global portfolios.

These findings indicate that portfolio risk is driven not only by individual asset volatility but also by macro-financial linkages and systemic dependencies.

7.6 Monte Carlo vs. Stress Testing

Monte Carlo simulation provides a forward-looking distribution of returns under modeled assumptions. However, its reliance on multivariate normality limits its ability to fully capture extreme events.

While simulated VaR estimates align broadly with historical values, stress testing reveals significantly larger losses under adverse conditions.

This confirms that:

simulation-based models must be complemented by scenario analysis to adequately capture real-world risk.

7.7 Key Takeaways

- **Portfolio risk is non-normal and asymmetric**
- **Tail events play a dominant role in loss generation**
- **Diversification is state-dependent and unreliable under stress**
- **VaR alone is insufficient for risk management**
- **Stress testing provides critical insight into extreme scenarios**

8. Conclusion

The analysis demonstrates that portfolio risk is fundamentally driven by tail events rather than average behavior, reinforcing the need for risk frameworks that extend beyond traditional distributional assumptions.

This project demonstrates a fully integrated stress testing framework combining:

- **historical risk analysis**
- **scenario-based shocks**
- **Monte Carlo simulation**

The results show that:

- **portfolios are highly vulnerable to systemic shocks**
- **diversification has limits**
- **traditional models underestimate extreme risk**

Combining multiple approaches provides a more robust and realistic assessment of portfolio risk.

9. Limitations & Future Work

Limitations

- **Normality assumption (underestimates tail risk)**
- **Static portfolio weights**
- **Reliance on historical data**

Future Improvements

- **Fat-tailed distributions (t-distribution)**
- **Volatility modeling (GARCH)**
- **Dynamic portfolio optimization**
- **Macro-driven stress scenarios**